EMERGING INFECTIONS

Emergence and spread of a human-transmissible multidrug-resistant nontuberculous mycobacterium

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Lung infections with *Mycobacterium abscessus*, a species of multidrug-resistant nontuberculous mycobacteria, are emerging as an important global threat to individuals with cystic fibrosis (CF), in whom *M. abscessus* accelerates inflammatory lung damage, leading to increased morbidity and mortality. Previously, *M. abscessus* was thought to be independently acquired by susceptible individuals from the environment. However, using whole-genome analysis of a global collection of clinical isolates, we show that the majority of *M. abscessus* infections are acquired through transmission, potentially via fomites and aerosols, of recently emerged dominant circulating clones that have spread globally. We demonstrate that these clones are associated with worse clinical outcomes, show increased virulence in cell-based and mouse infection models, and thus represent an urgent international infection challenge.

Nontuberculous mycobacteria (NTM; referring to mycobacterial species other than *Mycobacterium tuberculosis* complex and *Mycobacterium leprae*) are ubiquitous environmental organisms that can cause chronic pulmonary infections in susceptible individuals (*1*, *2*), particularly those with preexisting inflammatory lung diseases such as cystic fibrosis (CF) (*3*). The major NTM infecting CF individuals around the world is *Mycobacterium abscessus*; this rapidly growing, intrinsically multidrug-resistant species, which can be impossible to treat despite prolonged combination antibiotic therapy (*1*, *3*–*5*), leads to accelerated decline in lung function (*6*, *7*) and remains a contraindication to lung transplantation in many centers (*3*, *8*, *9*).

Until recently, NTM infections were thought to be independently acquired by individuals through exposure to soil or water (*10*–*12*). As expected, previous analyses from the 1990s and 2000s (*13*–*16*) showed that CF patients were infected with genetically diverse strains of *M. abscessus*, presumably from environmental sources. We used whole-genome sequencing at a single UK CF center and identified two clusters of patients (11 individuals in total) infected with identical or near-identical *M. abscessus* isolates, which social network analysis suggested were acquired within the hospital via indirect transmission between patients (*17*)—a possibility further supported through the genomic sequencing (*18*) of a separate *M. abscessus* outbreak in a Seattle, Washington, CF center (*19*).

Given the increasing incidence of *M. abscessus* infections in CF and non-CF populations reported globally (*3*, *20*, *21*), we investigated whether cross-infection, rather than independent environmental acquisition, might be the major source of infection for this organism; we therefore undertook population-level, multinational, whole-genome sequencing of *M. abscessus* isolates from infected CF patients, correlating results with clinical metadata and phenotypic functional analysis of isolates.

We generated whole-genome sequences for 1080 clinical isolates of *M. abscessus* from 517 patients, obtained from UK CF clinics and their associated regional reference laboratories, as well as CF centers in the United States [University of North Carolina (UNC), Chapel Hill], the Republic of Ireland (Dublin), mainland Europe (Denmark, Sweden, and Netherlands), and Australia (Queensland). We identified 730 isolates as *M. a. abscessus*, 256 isolates as *M. a. massiliense*, and 91 isolates as *M. a. bolletii*, with three isolates (from three different patients) containing more than one subspecies.

Phylogenetic analysis of these sequences (using one isolate per patient)—supplemented by published genomes from the United States, France, Brazil, Malaysia, China, and South Korea (table S1)—was performed and analyzed in the context of the geographical provenance of isolates (Fig. 1 and fig. S1). As done previously (*17*), we obtained maximum-likelihood phylogenetic trees demonstrating separation of *M. abscessus* into three clearly divergent subspecies (*M. a. abscessus*, *M. a. bolletii*, and *M. a. massiliense*), challenging recent reclassifications of *M. abscessus* into only two subspecies (*22*).

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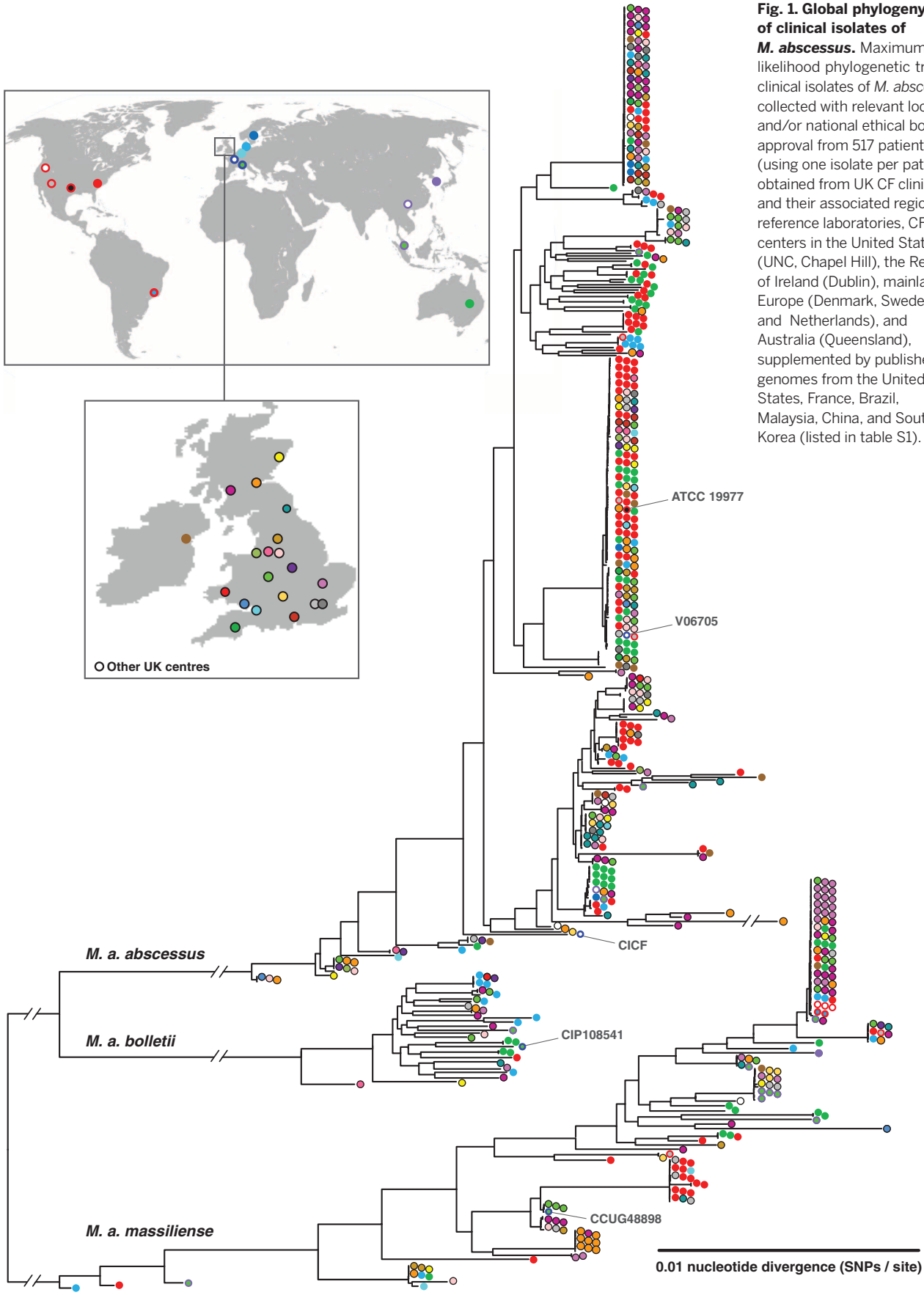

Fig. 1. Global phylogeny of clinical isolates of *M. abscessus*. Maximum likelihood phylogenetic tree of clinical isolates of *M. abscessus* collected with relevant local and/or national ethical board approval from 517 patients (using one isolate per patient), obtained from UK CF clinics and their associated regional reference laboratories, CF centers in the United States (UNC, Chapel Hill), the Republic of Ireland (Dublin), mainland Europe (Denmark, Sweden, and Netherlands), and Australia (Queensland), supplemented by published genomes from the United States, France, Brazil, Malaysia, China, and South Korea (listed in table S1).

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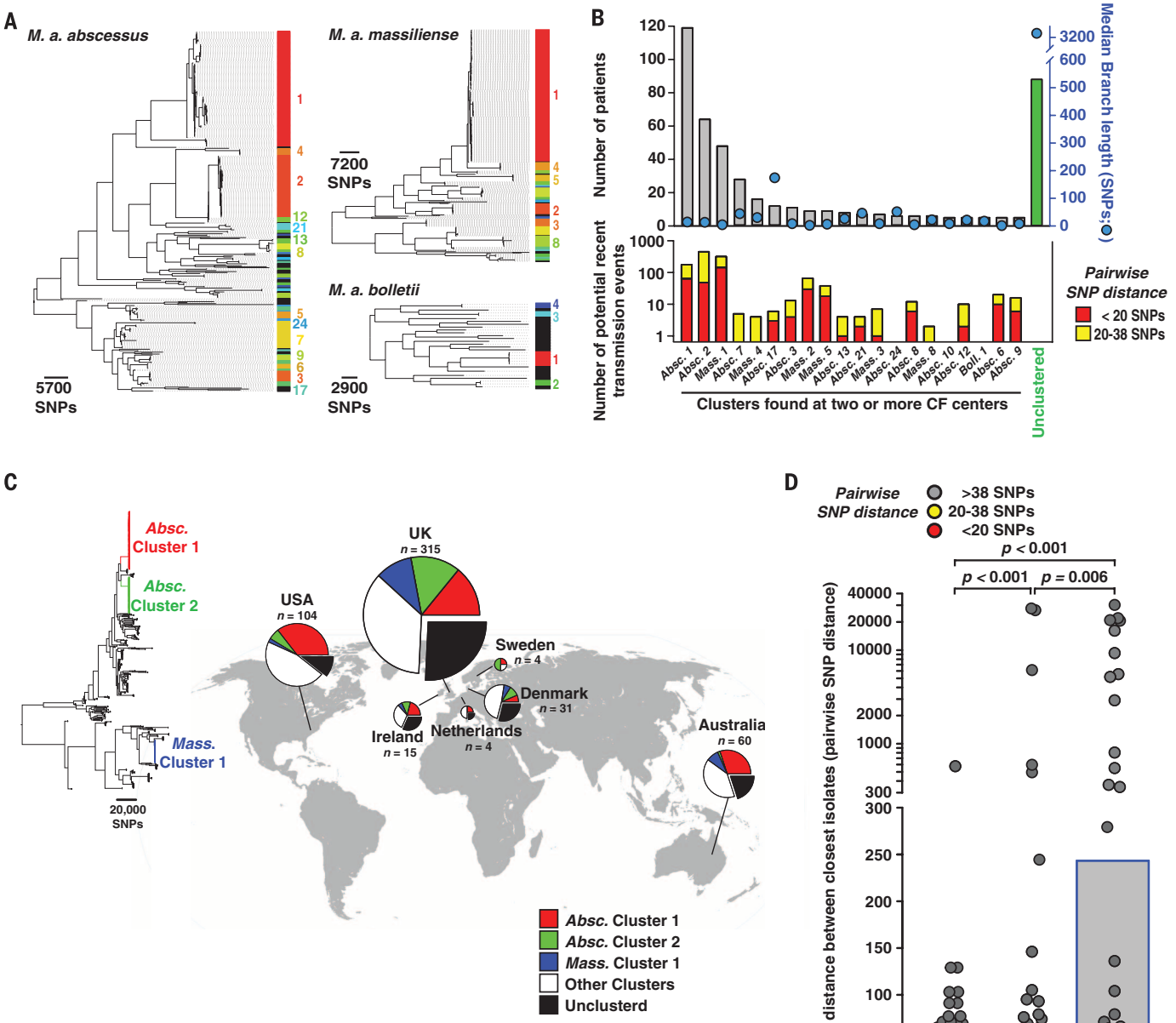

Fig. 2. Transcontinental spread of dominant circulating clones. (**A**) Hierarchical branch density analysis of phylogenetic trees for each subspecies of *M. abscessus* identifies multiple clusters of closely related isolates predominantly within the *M. a. abscessus* and *M. a. massiliense* subspecies (numbered, and spectrally colored red to blue, from most densely clustered to least; black indicates no significant clustering). (**B**) Analysis of *M. abscessus* clusters found in two or more CF centers showing (top) numbers of patients infected with each cluster (gray bars) or unclustered isolates (green) and median branch length (SNPs) of different patients' isolates within each cluster (blue circles). (Bottom) Numbers of potential recent transmission events with <20 SNPs (red) or 20 to 38 SNPs (yellow) difference between isolates from different patients. (**C**) Global distribution of clustered *M. abscessus* isolates showing *M. a. abscessus* cluster 1 (red) and cluster 2 (green), *M. a. massiliense* cluster 1 (blue), other clustered isolates (grouped together for clarity; white) and unclustered isolates (black) with numbers of patients (*n*) sampled per location. (**D**) Genetic differences between isolates (measured by pairwise SNP distance) from different patients attending the same CF center, different CF centers within the same country, or CF centers in different countries (boxes indicate median and interquartile range; *P* values were obtained from Mann Whitney Rank Sum tests). To exclude multiple highly distant comparisons, for each isolate only the smallest pairwise distance with an isolate from another patient is included. Color coding indicates whether there were <20 SNPs difference (red), 20 to 38 SNPs difference (yellow), or >38 SNPs difference (gray) between isolates from different patients.

Within each subspecies, we found multiple examples of deep branches (indicating large genetic differences) between isolates from different individuals, which is consistent with independent acquisition of unrelated environmental bacteria. However, we also identified multiple clades of near-identical isolates from geographically diverse locations (Fig. 1), suggesting widespread transmission of circulating clones within the global CF patient community.

To investigate further the relatedness of isolates from different individuals, we analyzed each subspecies phylogeny for the presence of high-density phylogenetic clusters (*23*). We identified multiple dense clusters of isolates, predominantly

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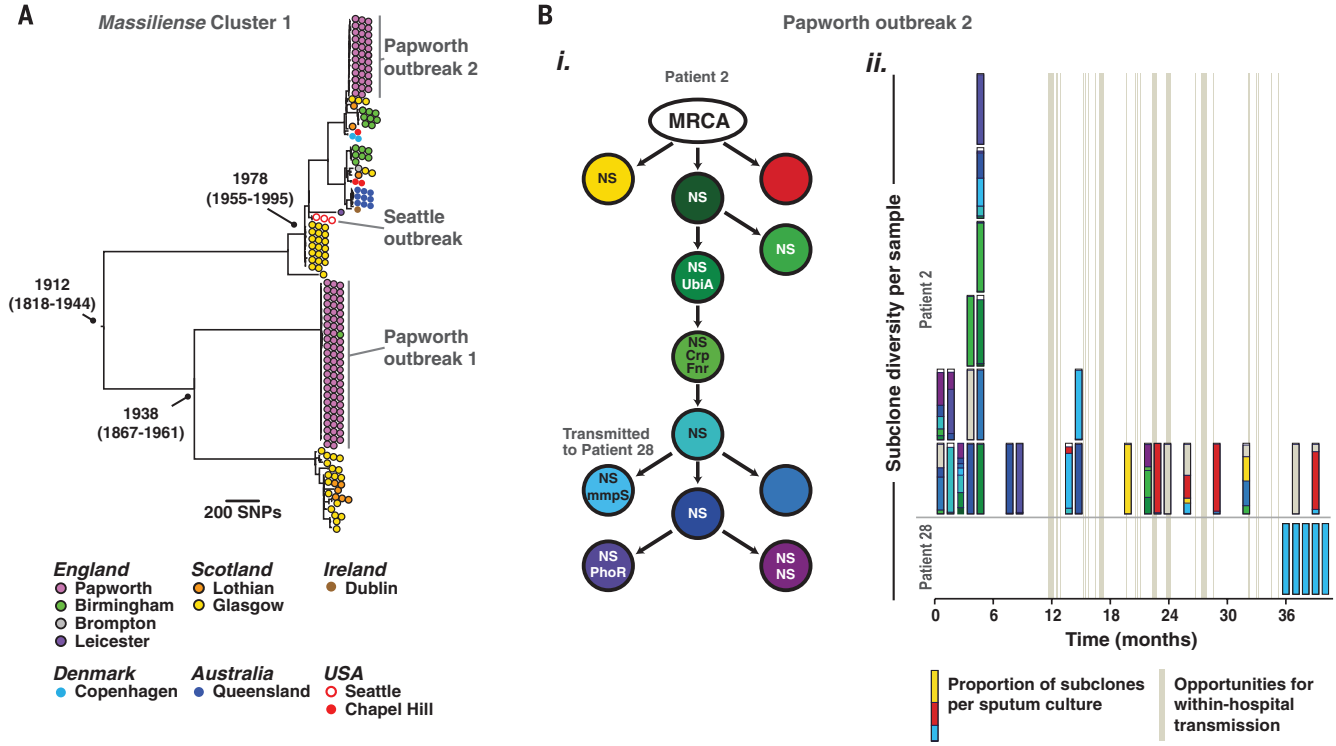

Fig. 3. Dating the emergence of dominant circulating clones. (**A**) Dating the emergence of the *M. a. massiliense* cluster 1 (responsible for the Papworth and Seattle CF center outbreaks), by using Bayesian analysis, with geographical annotation of isolates within the cluster. (**B**) Predicted evolution of subclones (identified through minority variant linkage) (*23*) within a single patient with CF [patient 2 from (*19*)] chronically infected with the dominant circulating clone *Massiliense* cluster 1 (representative of a total of 11 patients studied). (i) Analysis revealed successive acquisition of nonsynonymous polymorphisms (NS) by the most common recent ancestral clone (MRCA; white) in potential virulence genes (UBiA, MAB_0173; Crp/Fnr, MAB_0416c; mmpS, MAB_0477; and PhoR, MAB_0674) and then transmission of a single subclone to another patient from the same CF center [patient 28 from (*19*)]. (ii) Frequency of each subclone within longitudinal sputum isolates analyzed during the course of patient 2's infection and the subsequent transmission of a subclone to patient 28. We observed considerable heterogeneity in the detected repertoire of subclones within each sputum sample [vertical rectangles colored to illustrate the proportion of detected subclones coded as for (i) in each sputum sample], reflecting either temporal fluctuations in dominant sublineages or variable sampling of geographical diversity of subclones within the lung [as previously described for *P. aeruginosa* (*37*)]. Previously determined opportunities for hospital-based cross-infection between the two patients [using social network and epidemiologic analysis (*17*)] are shown in gray vertical bars.

within the *M. a. abscessus* and *M. a. massiliense* subspecies (Fig. 2A), indicating the presence of dominant circulating clones. We next excluded clusters found in only one CF center from further analysis so as to remove related isolates that might have been acquired from a local environmental point source. We found that most patients (74%) were infected with clustered, rather than unclustered, isolates, principally from *M. a. abscessus* cluster 1 and 2 and *M. a. massiliense* cluster 1 (Fig. 2B). The median branch lengths of almost all clusters found in two or more CF centers was less than 20 single-nucleotide polymorphisms (SNPs) (range of 1 to 175 SNPs), indicating a high frequency of identical or near identical isolates infecting geographically separate individuals.

To determine how much of the genetic relatedness found within clusters was attributable to recent transmission, we first examined the within-patient genetic diversity of *M. abscessus* isolates from single individuals. In keeping with our previously published results (*17*), we found that 90% of same-patient isolates differed by less than 20 SNPs, whereas 99% of same-patient isolates differed by less than 38 SNPs (fig. S2). We therefore classified isolates from different individuals varying by less than 20 SNPs as indicating "probable," and those varying by 20 to 38 SNPs as indicating "possible," recent transmission (whether direct or indirect). We thereby identified multiple likely recent transmission chains in virtually all multisite clusters of *M. abscessus* (Fig. 2B) and across the majority of CF centers (fig. S3).

We next examined the global distribution of clustered isolates and found that in all countries, the majority of patients were infected with clustered rather than unclustered isolates (Fig. 2C and table S2), suggesting frequent and widespread infection of patients with closely related isolates. Moreover, the three dominant circulating clones, *M. a. abscessus* clusters 1 and 2 and *M. a. massiliense* cluster 1, were all represented in the U.S., European, and Australian collections of clinical isolates, indicating transcontinental dissemination of these clades.

We then compared the genetic differences between isolates (measured by pairwise SNP distance) as a function of geography. As expected from our previous detection of hospital-based transmission of *M. abscessus* (*17*), average genetic distances were significantly shorter for *M. abscessus* isolates from the same CF center than those from different CF centers within the same country or from different countries (Fig. 2D). However, we also detected numerous examples of identical or near-identical isolates infecting groups of patients in different CF centers and, indeed, across different countries (Fig. 2D), indicating the recent global spread of *M. abscessus* clones throughout the international CF patient community.

We applied Bayesian analysis (*24*) to date the establishment and spread of dominant circulating clones (figs. S4 and S5), focusing on *M. a. massiliense* cluster 1, which includes isolates from both the Seattle (*19*) and Papworth (*17*) CF center outbreaks, as well as isolates from CF centers across England (Birmingham, London, and Leicester), Scotland (Lothian and Glasgow), Ireland (Dublin), Denmark (Copenhagen), Australia (Queensland), and the United States (Chapel Hill, North Carolina) (Fig. 3A). We estimate that the most recent common ancestor of isolates infecting patients from all these locations emerged around 1978

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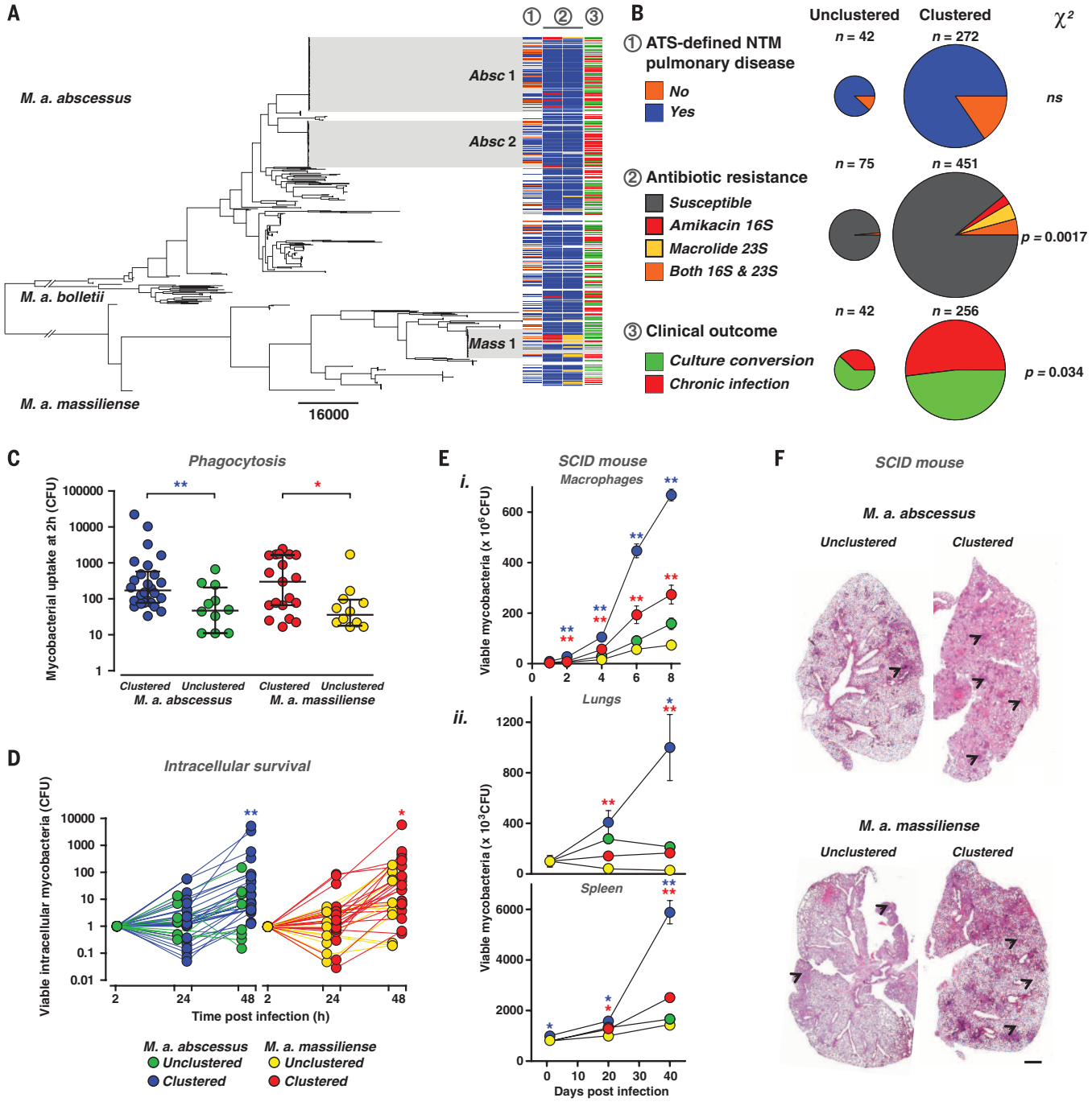

Fig. 4. Comparison of clinical outcomes and functional phenotyping of clustered and unclustered *M. abscessus* isolates. (**A** and **B**) Relationship of phylogeny with clinical metadata. (A) Phylogenetic tree of *M. abscessus* isolates (one isolate per patient) with dominant circulating clones *M. a. abscessus* 1 (Absc 1), *abscessus* 2 (Absc 2), and *M. a. massiliense* (Mass 1) highlighted (gray). (B) For each isolate, clinical data (where available) was used to determine whether (column 1) the infected patient fulfilled the ATS/Infectious Diseases Society of America (IDSA) criteria for NTM pulmonary disease, namely the presence of two or more culture-positive sputum samples with NTM-associated symptoms and radiological changes (*1*) (yes, blue; no, orange); whether (column 2) isolates have acquired amikacin resistance [through 16*S* ribosomal RNA (rRNA) mutations; red], macrolide resistance (through 23*S* rRNA mutations; yellow), or both [orange, (B) only]; and whether (column 3) patients culture converted (green) or remained chronically infected (red) with *M. abscessus*. (**C** and **D**). In vitro phenotyping of representative isolates of clustered (blue) and unclustered (green) *M. a. abscessus* and clustered (red) and unclustered (yellow) *M. a. massiliense*, comparing phagocytosis as shown in (C) and intracellular survival (normalized for uptake), as shown in (D), within differentiated THP1 cells. Data points represent averages of at least three independent replicates. (**E** and **F**) Using SCID mice, infection with clustered *M. a. abscessus* (blue) and *M. a. massiliense* (red) led to (E) greater intracellular survival within (i) bone marrow–derived macrophages in vitro and (ii) higher bacterial burdens in lung and spleen after in vivo inoculation with 1×10^7 bacilli per animal, with (F) worse granulomatous lung inflammation (arrowheads) than that of unclustered controls (*M. a. abscessus*, green; *M. a. massiliense*, yellow). Scale bar, 2 mm. Colony-forming unit (CFU) data are shown as mean ± SEM; $^*P < 0.05$; $^{**}P < 0.005$ (two-tailed unpaired Student's *t* test).

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(95% confidence interval, 1955–1995), clearly indicating recent global dissemination of this dominant circulating clone among individuals with CF (Fig. 3A).

Furthermore, we were able to resolve individual transmission events between patients infected with dominant circulating clones through two orthogonal approaches. First, using high-depth genomic sequencing of colony sweeps, we were able to track changes in within-patient bacterial diversity in sputum cultures of a single individual over time. By linking the frequency of occurrence of minority variants in longitudinal samples, we were able to define the presence of particular subclones within infected individuals, assign their likely evolutionary development (involving the successive acquisition of nonsynonymous mutations in likely virulence genes) (Fig. 3B), monitor their relative frequencies over time, and demonstrate their transmission between patients (Fig. 3B). Second, through longitudinal whole-genome sequencing of isolates collected over time from individuals, we were able to find multiple examples of the complete nesting of one patient's sampled diversity within another's (fig. S6). Such paraphyletic relationships are strongly indicative of recent person-to-person transmission (*25*).

We next examined potential mechanisms of transmission of *M. abscessus* between individuals [which our previous epidemiological analysis had suggested was indirect rather than via direct contact between patients (*17*)]. We provide proof of concept for fomite spread of *M. abscessus* (detecting three separate transmission events associated with surface contamination of an inpatient room by an individual infected with a dominant circulating clone) (fig. S7) and also for potential airborne transmission (by experimentally demonstrating the generation of long-lived, potentially infectious cough aerosols by an infected CF patient) (fig. S8).

A potential explanation for the emergence of dominant clones of *M. abscessus* is that they are more efficient at infection and/or transmission. We therefore analyzed clinical metadata to establish whether outcomes were different for patients infected with clustered rather than unclustered isolates. We correlated clinical outcomes with bacterial phylogeny and the presence of constitutive resistance to two key NTM antibiotics, amikacin and macrolides (*26*, *27*), acquired through point mutations in the 16*S* and 23*S* ribosomal RNA, respectively (Fig. 4A). We found no differences in the proportions of *M. abscessus*–positive individuals diagnosed with American Thoracic Society (ATS)–defined NTM pulmonary disease (namely, the presence of two or more culture-positive sputum samples with NTM-associated symptoms and radiological changes) (*1*), but we did observe increased rates of chronic infection in individuals infected with clustered rather than unclustered isolates (Fig. 4B). As anticipated for transmissible clones exposed to multiple rounds of antibiotic therapy, we also found high rates of constitutive amikacin and/or macrolide resistance in clustered isolates (Fig. 4B). Resistance to these two antibiotics did not necessarily result in poor clinical outcomes (fig. S9), suggesting that additional bacterial factors might contribute to worse responses in patients infected with clustered isolates.

To explore differences in intrinsic virulence between clustered and unclustered *M. abscessus*, we subjected a panel of representative isolates (27 clustered and 17 unclustered *M. a. abscessus*; 25 clustered and 13 unclustered *M. a. massiliense*) to a series of in vitro phenotypic assays. Although we found no or only minor differences between groups in their colony morphotype, biofilm formation, ability to trigger cytokine release from macrophages (fig. S10), and their overall phenotypic profile (determined with multifactorial analysis) (fig. S11), we detected significantly increased phagocytic uptake (Fig. 4C) and intracellular survival in macrophages (Fig. 4D) of clustered isolates of both *M. a. abscessus* and *M. a. massiliense* as compared with unclustered controls, indicating clear differences in pathogenic potential. Moreover, infection of severe combined immunodeficient (SCID) mice revealed significantly greater bacterial burden (Fig. 4E) and granulomatous inflammation (Fig. 4F) after inoculation with clustered rather than unclustered isolates of *M. a. abscessus* and *M. a. massiliense*, confirming differences in virulence between these groups.

Our results reveal that the majority of *M. abscessus* infections of individuals with CF worldwide are caused by genetically clustered isolates, suggesting recent transmission rather than independent acquisition of genetically unrelated environmental organisms. Given the widespread implementation of individual and cohort segregation of patients in CF centers in Europe (*28*), the United States (*29*), and Australia (*30*) [which have led to falling levels of methicillin-resistant *Staphylococcus aureus* (MRSA), *Burkholderia*, and transmissible *Pseudomonas* infections (*31*–*33*)], we believe that the likely mechanism of local spread of *M. abscessus* is via fomite spread or potentially through the generation of long-lived infectious aerosols [as identified for other CF pathogens (*34*–*36*)]. Although further research is needed, both transmission routes are plausible given our findings (figs. S7 and S8) and would be potentially enhanced by the intrinsic desiccation resistance of *M. abscessus*. Such indirect transmission, involving environmental contamination by patients, is supported by our previous social network analysis of a UK outbreak of *M. abscessus* (*17*) in CF patients, which revealed hospital-based cross-infection without direct person-to-person contact, and by the termination of a Seattle *M. abscessus* outbreak associated with the introduction of clinic room negative pressure ventilation and double room cleaning (*19*). The long-distance spread of circulating clones is more difficult to explain. We found no evidence of CF patients or of equipment moving between CF centers in different countries, indicating that the global spread of *M. abscessus* may be driven by alternative human, zoonotic, or environmental vectors of transmission.

Our study illustrates the power of population-level genomics to uncover modes of transmission of emerging pathogens and has revealed the recent emergence of global dominant circulating clones of *M. abscessus* that have spread between continents. These clones are better able to survive within macrophages, cause more virulent infection in mice, and are associated with worse clinical outcomes, suggesting that the establishment of transmission chains may have permitted multiple rounds of within-host genetic adaptation to allow *M. abscessus* to evolve from an environmental organism to a true lung pathogen.

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SUPPLEMENTARY MATERIALS

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NEUROSCIENCE

Neural correlates of ticklishness in the rat somatosensory cortex

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Rats emit ultrasonic vocalizations in response to tickling by humans. Tickling is rewarding through dopaminergic mechanisms, but the function and neural correlates of ticklishness are unknown. We confirmed that tickling of rats evoked vocalizations, approach, and unsolicited jumps (Freudensprünge). Recordings in the trunk region of the rat somatosensory cortex showed intense tickling-evoked activity in most neurons, whereas a minority of cells were suppressed by tickling. Tickling responses predicted nontactile neural responses to play behaviors, which suggests a neuronal link between tickling and play. Anxiogenic conditions suppressed tickling-evoked vocalizations and trunk cortex activity. Deep-layer trunk cortex neurons discharged during vocalizations, and deep-layer microstimulation evoked vocalizations. Our findings provide evidence for deep-layer trunk cortex activity as a neural correlate of ticklishness.

Tickling sensations can be differentiated into laughter-inducing "gargalesis" and non–laughter-inducing light touch, "knismesis" (*1*). The former is a peculiar, often funny form of social touch, which has been discussed for more than two millennia (*2*, *3*). Still, major questions remain unanswered: Why does tickling induce laughter? Why are tickling effects so mood-dependent (*4*)? Why do body parts differ in ticklishness (*1*)? Why can't we tickle ourselves (*2*)? Is ticklish laughter different from humorous laughter (*4*–*6*)? To address such questions, we need a better understanding of the neural correlates of ticklishness. We took advantage of groundbreaking advances that provided evidence for tickling-evoked 50-kHz vocalizations and primitive forms of joy in rats (*7*, *8*). On the basis of this work, we measured "rat ticklishness" in our work as the propensity to "laugh" (emit 50-kHz calls) upon being tickled. We focused on the somatosensory cortex because it is the largest tactile neural representation in mammals, because human imaging studies suggested this candidate region (*9*, *10*), and because work on somatosensory afferents provided no conclusive evidence for dedicated peripheral mechanisms of tickle.

We tickled and gently touched rats on different body parts (Fig. 1A, ventral tickling) and observed a variety of ultrasonic vocalizations (USVs; Fig. 1B), in particular during tickling (movie S1). Rat 50-kHz vocalizations indicate positive emotional valence (*11*). Consistent with earlier claims that tickling is rewarding through the dopaminergic system (*7*, *12*, *13*), rats rapidly approached the tickling hand, and tickling induced unsolicited jumps accompanied by 50-kHz USVs (Freudensprünge, "joy jumps"; movie S2), which can be seen in joyful subjects in various mammalian species (*14*–*16*). We visually categorized spectrograms of an extensive set of USVs (34,140 calls) into modulated, trill, combined, and miscellaneous call types (Fig. 1B and fig. S1). Both tickling and gentle touch evoked USVs (Fig. 1C), but tickling induced more USVs than did gentle touch (ventral tickling, 4.45 ± 0.28 Hz; ventral gentle touch, 2.58 ± 0.21 Hz; $n = 16$, $P < 0.001$; mean ± SEM, paired t test). Rats seemed to warm up to tickling and vocalized less before the initial interaction than during breaks between interaction episodes (Pre versus Break; Fig. 1, C and D). Tickling the ventral trunk evoked the largest number of USVs (Fig. 1D) and the largest fraction of combined USVs (Fig. 1E). Play behavior (rat chasing experimenter's hand; Fig. 1F) also evoked USVs (Fig. 1, G and H, and movie S3) (*17*). Consistent with our sense that rats experienced the experimental setting as emotionally positive, we did not observe 22-kHz alarm calls.

We simultaneously performed single-unit recordings in the trunk region of the somatosensory cortex (Fig. 2A). We obtained high-quality recordings of neuronal responses elicited by tickling and gentle touch (Fig. 2B and fig. S2A). Similar to USVs, activity in the trunk cortex was lower before initial tickling (Fig. 2, B and C, Pre) than during the short breaks between interactions (Fig. 2, B and C, Break). Remarkably, neuronal responses were also observed during hand-chasing phases, when rats were not touched by the experimenter (Fig. 2B and fig. S2E). Most cells increased their firing rate during trunk tickling, trunk gentle touch, and chasing hand (~77%, ~67%, and ~80% of the cells showed higher firing rates during interaction than during break, respectively; fig. S2B, top, and C to E), whereas a minority of cells were suppressed during interaction phases (fig. S2B, bottom, and C to E). Similar to USVs, neuronal firing rates increased more during tickling than during gentle touch on the trunk (Fig. 2D). As in the cells shown in fig. S2B, responses to tickling predicted play responses (chasing hand) across the population (Fig. 2E), which suggests a neural link between tickling and play behavior.

Anxiogenic conditions suppress tickling-evoked USVs in rats (*7*). To test whether neuronal responses to tickling are also modulated by such conditions, we tickled rats in both control (Fig. 3A, left) and anxiogenic settings, such as under bright illumination and on an elevated platform (Fig. 3A, right). Tickling-evoked USVs were significantly suppressed in the anxiogenic condition and recovered in control conditions (Fig. 3, B and C). Similarly, anxiogenic conditions suppressed neuronal response to tickling (Fig. 3D) and inverted the sign of response index to tickling (Fig. 3E).

Our recordings revealed that USVs and neuronal activity in the trunk cortex are modulated in a similar way by tickling and anxiogenic conditions. We wondered whether tickling-evoked USVs and neuronal responses to tickling are causally linked. We therefore aligned neuronal firing to the onsets of USVs (Fig. 4, A and B). To avoid a confounding effect of the coactivation of the trunk cortex and USVs by tickling and touch, we restricted this analysis to break periods. The activity of trunk somatosensory

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Emergence and spread of a human-transmissible multidrug-resistant nontuberculous mycobacterium

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Global spread of aggressive mycobacteria

Many mycobacteria, in addition to those causing leprosy and tuberculosis, are capable of infecting humans. Some can be particularly dangerous in patients suffering from immunosuppression or chronic disease, such as cystic fibrosis (CF). Bryant *et al.* observed clusters of near-identical isolates of drug-resistant *Mycobacterium abscessus* in patients reporting to CF clinics. The similarity of the isolates suggests transmission between patients, rather than environmental acquisition. Although this bacterium is renowned for its environmental resilience, the mechanism for its long-distance transmission among the global CF patient community remains a puzzle.

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